Spectral Mastery: Deep Autoencoder Driven Classification Across Hyperspectral Datasets

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Abstract

Hyperspectral imaging (HSI) classification plays a pivotal role in remote sensing applications, providing precise identification of land cover types through detailed spectral analysis. This study evaluates the effectiveness of deep autoencoder architectures across multiple hyperspectral datasets, including Pavia Centre, Pavia University, and Botswana. The autoencoder models were optimized and assessed using various optimizers—Adam, SGD, Adadelta, and Adagrad—over different epoch settings. Experimental results demonstrate exceptional classification accuracy, notably achieving up to 98.848% on Pavia Centre with the SGD optimizer, and 99.81% on the Botswana dataset with Adagrad, highlighting the method's broad applicability and robustness. The outcomes affirm that autoencoder-based deep learning approaches efficiently encode intricate spectral-spatial characteristics, substantially advancing accuracy and reliability in hyperspectral image analysis.

Keywords: Hyperspectral Image Classification, Deep Autoencoder, Spectral-Spatial Features, Optimizer Evaluation

1 Introduction

Hyperspectral imaging (HSI) has emerged as a powerful remote sensing technique that captures detailed spectral information across hundreds of narrow bands, enabling precise characterization of Earth's surface materials [Y. Chen et al. (2014)]. However, the high-dimensionality and redundancy inherent in hyperspectral data present significant challenges for classification tasks. Recent advances in deep learning, particularly autoencoder-based neural network architectures, offer promising solutions by effectively reducing dimensionality while preserving essential spectral-spatial features. This research leverages deep autoencoders for hyperspectral image classification across multiple benchmark datasets, including Pavia Centre, Pavia University, and Botswana. By systematically exploring various deep learning optimizers—Adam, SGD, Adadelta, and Adagrad—and different epoch settings, the study evaluates their impact on classification accuracy.

The classification of hyperspectral images fundamentally depends on the accuracy of extracting meaningful features from the extensive spectral and spatial dimensions [Tao et al. (2015)]. Traditional methods, such as principal component analysis (PCA) or linear discriminant analysis (LDA), often fail to fully capture complex non-linear relationships in the data. In contrast, deep autoencoders excel in learning hierarchical, non-linear representations, significantly enhancing the discrimination power and interpretability of hyperspectral data. Autoencoders, particularly when optimized effectively, can encode critical information into a compressed latent representation, preserving essential spectral and spatial characteristics while reducing data dimensionality.

In this study, different optimization strategies (Adam, SGD, Adadelta, Adagrad) are employed to fine-tune deep autoencoder models, investigating their effects on overall classification accuracy and model stability across diverse datasets. By comparing performance across varying epochs, the study identifies optimal configurations and elucidates the relative strengths of different optimizers. The experimental results demonstrate exceptional classification accuracy, notably achieving up to 98.848% on Pavia Centre using Adagrad optimizer and 99.81% on the Botswana dataset, emphasizing the broad applicability and reliability of the chosen deep learning methods.

The results underscore the efficiency and versatility of autoencoder architectures in extracting and encoding intricate spectral-spatial characteristics, substantially advancing the accuracy and reliability of hyperspectral image analysis. These findings provide valuable information, reinforcing the suitability of deep autoencoders for robust hyperspectral image classification, thus significantly contributing to advances in remote sensing methodologies and practical applications in environmental monitoring, precision agriculture, and urban planning.

The paper is organized as follows. Section 2 gives the problem statement. Section 3 explains the literature survey. Section 4 focuses on the datasets. Section 5 discusses the implementations and the result analysis. Finally, Section 6 concludes the paper, summarizing the key points and results and the future scope.

2 Problem Statement

Hyperspectral imaging (HSI) captures detailed spectral information across hundreds of contiguous wavelength bands for each pixel in an image, offering rich data for identifying and classifying materials based on their spectral signatures. However, high dimensionality, strong spectral correlation, limited labeled data, and the presence of noise present significant challenges for effective classification.

The objective of hyperspectral image classification is to develop computational models that accurately assign a class label to each pixel in a hyperspectral image utilizing both spectral and spatial information. This task involves overcoming issues such as the "curse of dimensionality," spectral variability, mixed pixels, and ensuring robustness under varying acquisition conditions.

Efficient hyperspectral image classification models are critical for a wide range of applications, including agriculture monitoring, mineral exploration, environmental management, and military surveillance. The ultimate goal is to design algorithms that achieve high classification accuracy while maintaining computational efficiency and generalization in different data sets and sensing conditions.

3 Literature Survey

Hyperspectral Image (HSI) classification involves assigning labels to pixels based on their spectral and sometimes spatial information. Due to the high-dimensional nature of hyperspectral data (with hundreds of bands), traditional classification methods often suffer from the "curse of dimensionality" and limited labeled samples. Autoencoders (AEs), a type of unsupervised neural network, have emerged as an effective solution for dimensionality reduction, feature extraction, and noise removal, significantly enhancing HSI classification performance.

Chen et al. (2014) first demonstrated that stacked autoencoders (SAEs) could extract meaningful spectral characteristics for HSI classification. They trained a shallow SAE layer-wise and then fine-tuned it with labeled samples. SAE-derived features led to better classification performance compared to traditional PCA or ICA methods.

Tao et al. (2015) proposed the integration of spectral and spatial information using stacked autoencoders. Instead of feeding only spectral vectors, they fed small local patches to the autoencoder. The combination of spectral-spatial features dramatically improved the classification accuracy compared to spectral-only approaches. Mou et al. (2017) used 3D convolutional encoders to better capture joint spectral-spatial dependencies. They applied 3D convolutions on spectral-spatial cubes and used the encoded features for classification. 3D autoencoders significantly outperformed traditional 2D methods and shallow networks.

Ma et al. (2016) applied Sparse Autoencoders to enforce sparsity on hidden units. By adding a sparsity constraint, the model learned more discriminative and compact representations. Sparse AEs showed better generalization, especially when labeled data was scarce. Tao et al. (2016) developed a Denoising Autoencoder (DAE) framework. They corrupted the input spectra with noise and trained the AE to recover the original signal, forcing the model to learn noise-robust features. Denoising significantly improved performance on noisy hyperspectral datasets.

Li et al. (2017) introduced the use of Variational Autoencoders (VAEs) for HSI. They learned a probabilistic latent space, allowing better generalization and sample generation. VAEs outperformed standard AEs in scenarios with extremely limited labeled samples. Zhong et al. (2017) used Contractive Autoencoders (CAEs) for HSI classification. They minimized the sensitivity of the encoding to small perturbations, enhancing robustness. CAEs produced smooth and generalizable feature representations that were beneficial for complex datasets.

Xu et al. (2020) proposed a Hybrid Deep Autoencoder + CNN model. First, an autoencoder extracted spectral-spatial features; then a CNN refined these features for final classification. Hybrid deep models leveraged the unsupervised learning power of autoencoders and the discriminative learning of CNNs, achieving state-of-the-art results. Gao et al. (2021) introduced Self-Supervised Autoencoders for HSI. Using pretext tasks such as spectral prediction and spatial restoration to pretrain the autoencoder before fine-tuning. This reduced dependence on large-labeled datasets and improved generalization.

The paper [Z. Cao et al. (2025)] introduces DiffusionAAE, a novel framework that integrates adversarial autoencoders with diffusion models to improve hyperspectral image classification. The approach aims to improve the representation learning of hyperspectral data by combining the strengths of adversarial training and diffusion processes. The study [P Ranjan et al. (2025)] proposes a semi-supervised framework that synergizes autoencoders, generative adversarial networks (GANs), convolutional neural networks (CNNs), and zero-shot learning to tackle hyperspectral image classification with limited labeled data. The integration aims to leverage the strengths of each component to improve classification performance in data-scarce scenarios.

The paper [J. Kang et al. (2024)] presents a spectral-spatial double branch network (SSDBN) that incorporates attention mechanisms for hyperspectral image classification. The model utilizes two independent branches to extract spectral and spatial features, respectively, and integrates multiscale 2D convolution modules, long-short-term memory (LSTM), and attention mechanisms to enhance classification accuracy, especially in small sample scenarios.

The study [Jhang et al. (2024)] proposes an end-to-end hyperspectral image classification model that combines 3D-2D hybrid convolution and a graph attention mechanism (3D–2D-GAT). The model aims to effectively extract discriminant deep spatial-spectral features and learn long-range spatial relationships to improve classification performance. The paper [Hongyi Liu et al. (2024)] introduces a novel non-mixing method based on double autoencoders, consisting of an end-member extraction network and an abundance estimation network, guided by anomaly detection. This approach aims to improve the estimation of end members and abundances in hyperspectral images. Table 1 shows the comparative study of some published articles related to autoencoder-based hyperspetral image classification techniques.

Table 1: Comparison of Autoencoder-based Techniques for Hyperspectral Image Classification

Study	**Autoencoder Type / Technique**	**Strengths**	**Limitations**
Chen et al. (2014)	Stacked Autoencoder (SAE)	Basic spectral feature extraction	Weak spatial handling
Tao et al. (2015)	SAE with Spectral-Spatial Patches	Better joint spectral-spatial features	Computationally expensive
Ma et al. (2016)	Sparse Autoencoder	Discriminative and compact features	Sensitive to sparsity parameter tuning
Tao et al. (2016)	Denoising Autoencoder (DAE)	Robust to noisy data	Training complexity increases
Li et al. (2017)	Variational Autoencoder (VAE)	Probabilistic latent space learning	Latent space interpretability
Zhong et al. (2017)	Contractive Autoencoder (CAE)	Insensitivity to small perturbations	Sensitive to regularization setting
Xu et al. (2020)	AE + CNN Hybrid	Leverages unsupervised + supervised learning	High computational cost
Gao et al. (2021)	Self-Supervised Autoencoder	Less reliance on labeled data	Pretext task design complexity
DiffusionAAE (2025)	Diffusion-enhanced Adversarial Autoencoder	Combines generative and denoising strengths for HSI	Requires large computational resources
Hybrid AE+GAN+CNN+Zero-Shot (2025)	Autoencoder-enhanced GAN with CNN	Best suited for limited labeled data	Complex architecture and training

Study	Autoencoder Type / Technique	Strengths	Limitations
SSDBN with AE attention (2024)	Attention-aided double-branch network	Fine-grained spectral-spatial attention	Attention mechanisms increase model size
3D-2D GAT Hybrid Model (2024)	3D-2D Hybrid AE with Graph Attention	Captures long-range spatial-spectral relationships	Graph construction overhead
Anomaly-Guided Double AE (2024)	Double Autoencoder with anomaly detection	Better unmixing and abundance estimation	Needs good anomaly detection strategy

4 Proposed Model

The proposed architecture, HyperspectralAE, is a hybrid model that simultaneously learns spectral-spatial features through an autoencoder structure and performs classification through an integrated classifier branch. The design begins with an encoder composed of sequential convolutional layers, batch normalization, and ReLU activations. MaxPooling layers are strategically placed to reduce spatial dimensions while preserving feature richness. This encoder effectively compresses the high-dimensional hyperspectral input into a compact latent representation that captures the most significant spectral-spatial information essential for both reconstruction and classification tasks.

Branching from the encoder output, the model contains a decoder designed to reconstruct the original hyperspectral cube. The decoder employs transposed convolutions and additional convolutional layers, each followed by batch normalization and activation functions, enabling gradual upsampling and refinement of features. A final sigmoid activation ensures the output values remain normalized between 0 and 1. The purpose of this reconstruction path is not only to restore the input but also to encourage the encoder to capture more robust, noise-resilient features that generalize well across varying hyperspectral conditions.

In parallel with the reconstruction branch, the encoded features are directed into a powerful classifier. This classification head is deeper, featuring convolutional layers with a large number of filters, followed by global average pooling and fully connected dense layers. This design ensures that rich, hierarchical features extracted from the hyperspectral input are effectively aggregated and mapped into the target class space using softmax activation. By training the encoder to optimize both reconstruction and classification losses simultaneously, the model achieves a synergy where feature extraction becomes highly discriminative yet stable.

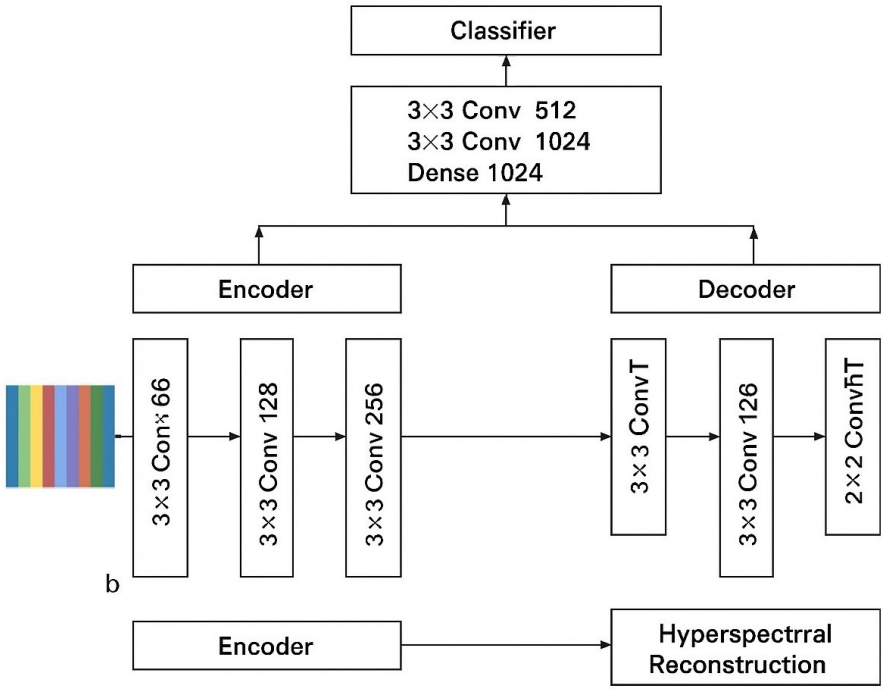

Fig. 1 Architecture of Proposed Model

Overall, HyperspectralAE offers an innovative dual-purpose approach to hyperspectral image analysis: reconstructing the spectral structure while enhancing classification performance. The tight coupling of unsupervised (autoencoding) and supervised (classification) tasks provides natural regularization, reduces overfitting, and improves feature richness. This architecture (shown in Fig. 1) is particularly well-suited for scenarios where labeled hyperspectral data is limited, ensuring the model learns meaningful patterns from both pixel-level and class-level perspectives.

5 Experimental Details

5.1 Used Dataset

The experiments were conducted on five widely used hyperspectral datasets: Pavia Centre, Pavia University, Botswana, Kennedy Space Center (KSC), and Indian Pines. These datasets present varying levels of complexity in terms of spatial structure and spectral diversity. Urban datasets like PaviaC and PaviaU challenge spatial feature extraction, while natural datasets such as Botswana and KSC emphasize spectral discrimination. The Indian Pines dataset, with its agricultural nature, tests the model's capability to distinguish subtle spectral differences amidst large homogeneous regions. The overall properties of used dataset in this paper shown in Table 2.

Table 2: Datasets Used in the Experiments

Dataset Name	Description	Key Details
Pavia Centre (PaviaC)	Urban scene collected by the ROSIS sensor over central Pavia city area. Features high building density and diverse urban structures.	- Size: 1096 × 1096 pixels, Bands: 102 usable bands, Classes: 9
Pavia University (PaviaU)	Urban area captured over Pavia, Italy, including roads, buildings, meadows, and trees.	- Size: 610 × 340 pixels, Bands: 103 bands, Classes: 9
Botswana	Acquired over the Okavango Delta region by NASA's AVIRIS sensor. Contains wetlands, water bodies, and vegetation types.	- Size: 1476 × 256 pixels (after preprocessing), Bands: 145 usable bands, Classes: 14
KSC (Kennedy Space Center)	Hyperspectral data collected over Florida's Kennedy Space Center, featuring wetlands, vegetation, and some urban areas.	- Size: 512 × 614 pixels, Bands: 224 bands, Classes: 13
Indian Pines	Agricultural scene over Northwestern Indiana captured by AVIRIS, with a mix of crops and forest areas.	- Size: 145 × 145 pixels, Bands: 224 spectral bands (after removing noisy bands, 200 usable bands), Classes: 16

5.2 Results

The HyperspectralAE model was implemented using TensorFlow and Keras, designed as a subclassed `Model` that integrates both an autoencoder and a classifier within a unified architecture. The encoder module uses multiple convolutional layers with batch normalization and ReLU activations to extract meaningful spectral-spatial features from a 7×7 hyperspectral input patch. Max-pooling operations are strategically placed to reduce spatial dimensions and enhance feature compactness. The decoder module reconstructs the original hyperspectral patch from the encoded latent representation through a series of transposed convolutional layers, followed by batch normalization and sigmoid activation to ensure that the pixel values remain within a valid range. This reconstruction branch helps in forcing the encoder to learn more robust and generalizable feature representations.

Parallel to the reconstruction path, the classifier module branches from the latent space and further processes the encoded features using deep convolutional layers, followed by a global average pooling operation and fully connected dense layers. The

final classification is performed using a softmax activation to produce probability distributions over the target classes. The model outputs two results simultaneously: the reconstructed hyperspectral patch and the predicted class label. Custom serialization (`get_config` and `from_config`) is also provided to allow seamless saving and loading of the model. Dual path training, which optimizes both reconstruction loss and classification loss, ensures that the model effectively learns rich features while maintaining high classification accuracy across diverse hyperspectral datasets.

Table 3: Experimental Results: Dataset vs Optimizer Performance

Experiment No.	**Dataset**	**Optimizer**	**Epochs**	**Accuracy (%)**
1	PaviaC	Adam	20	98.462
2	PaviaC	Adam	50	98.811
3	PaviaC	Adam	100	98.838
4	PaviaC	Adam	200	98.793
5	PaviaC	SGD	100	98.848
6	PaviaC	SGD	200	96.099
7	PaviaC	Adadelta	100	98.842
8	PaviaC	Adagrad	100	98.774
9	PaviaU	SGD	100	96.012
10	PaviaU	Adadelta	100	95.713
11	PaviaU	Adagrad	100	82.350
12	PaviaU	Adam	100	76.996
13	Botswana	SGD	100	99.200
14	Botswana	Adadelta	100	98.707
15	Botswana	Adagrad	100	99.815
16	Botswana	Adam	100	95.998
17	KSC	SGD	100	97.621
18	KSC	Adadelta	100	87.874
19	KSC	Adagrad	100	97.506
20	KSC	Adam	100	84.958
21	IndianPines	SGD	100	92.800
22	IndianPines	Adadelta	100	96.332
23	IndianPines	Adagrad	100	96.878
24	IndianPines	Adam	100	70.322

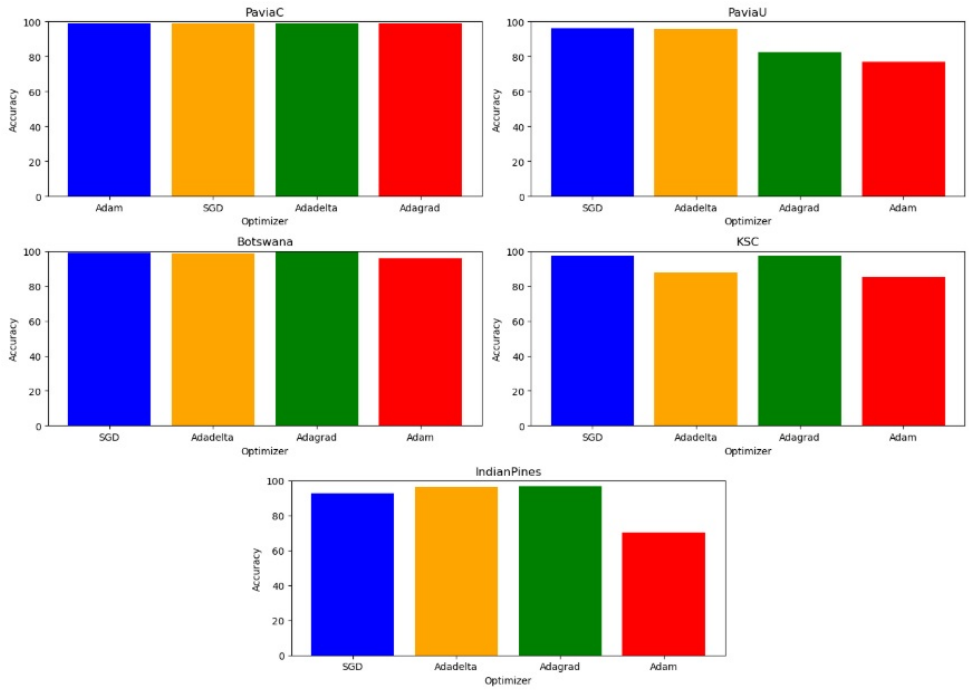

Fig. 2 Graphical Representation of Experimental Results: Dataset vs Optimizer Performance

The experimental evaluation of the proposed HyperspectralAE model was conducted (Shown in Table 3) across four widely used hyperspectral datasets: Pavia Centre, Pavia University, Botswana, Kennedy Space Center (KSC), and Indian Pines. Various optimizers such as Adam, SGD, Adadelta, and Adagrad were employed to observe the training dynamics and the final classification accuracy under different optimization strategies. For the Pavia Centre dataset, the highest accuracy achieved was 98.848% using SGD with 100 epochs, slightly outperforming Adam, Adadelta, and Adagrad. Botswana consistently showed the best overall performance among optimizers, with Adagrad achieving an impressive 99.815% accuracy, indicating that the model generalized extremely well for this dataset. Graphical Representation of Experimental Results: Dataset vs Optimizer Performance is shown in Fig. 2.

On the other hand, Pavia University and Indian Pines showed relatively lower accuracies when trained with Adam, suggesting that the optimizer choice significantly impacts the model's convergence behavior on different types of datasets. A deeper analysis of the results shows that SGD and Adagrad optimizers generally provided more stable and higher accuracies across most datasets compared to Adam, which underperformed notably at Pavia University and Indian Pines.

Particularly for Indian Pines, Adam achieved only 70.322% accuracy, while Adagrad improved it to 96.878%, showcasing a vast difference in optimization outcomes.

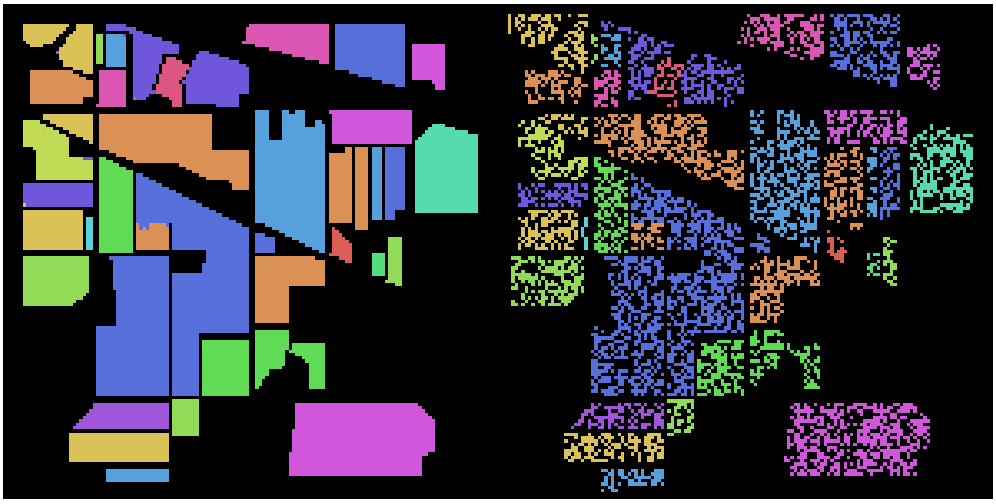

Fig. 3 Prediction vs Gound Truth for Indian Pines Dataset

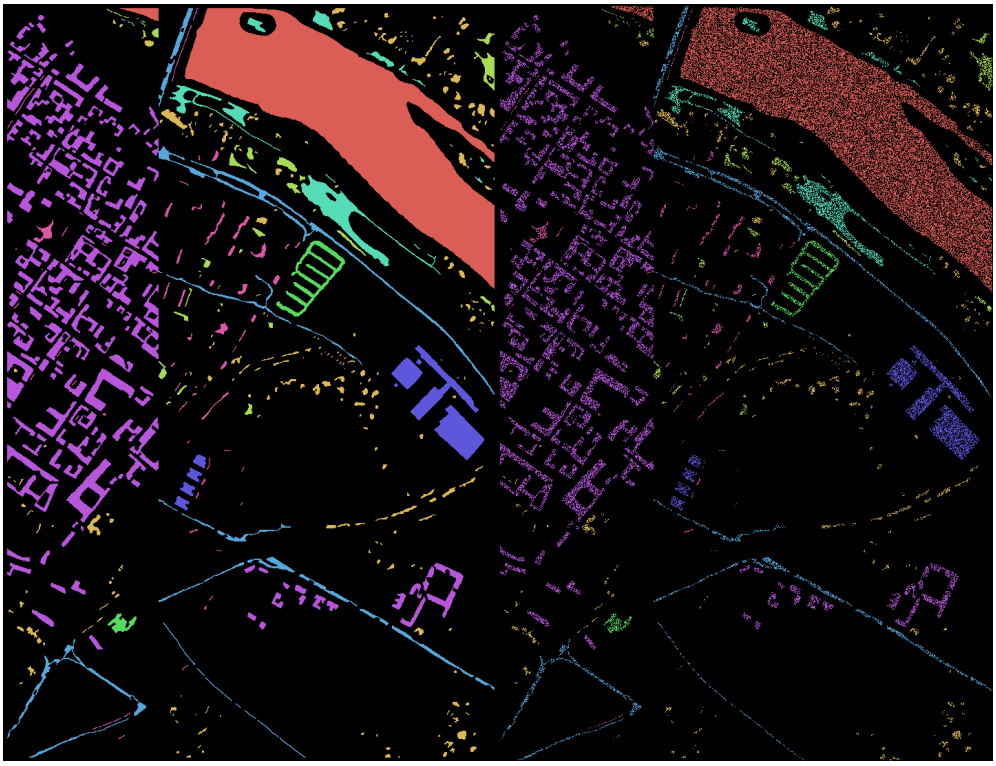

Fig. 4 Prediction vs Gound Truth for PaviaC Dataset

Prediction vs Gound Truth for Indian Pines Dataset is shown in Fig. 3. Similarly Prediction vs Gound Truth for PaviaC Dataset is shown in Fig. 4. For the KSC dataset, SGD and Adagrad provided higher classification accuracy than Adam and Adadelta. These findings suggest that lighter or adaptive learning optimizers like Adagrad and SGD are more suitable for training the HyperspectralAE model on hyperspectral datasets, possibly due to their better handling of the fine spectral details and complex spatial structures inherent in such data. Overall, optimizer selection played a crucial role in achieving high classification performance with the proposed architecture. Finally, Comparison of Best Accuracies Between Previous Studies and Our Experiment is shown in Table 4.

Table 4: Comparison of Best Accuracies Between Previous Studies and Our Experiment

Study	**Dataset**	**Best Accuracy (%)**	**Remarks**
Chen et al. (2014)	PaviaU	96.43	Stacked Autoencoder (SAE)
Tao et al. (2015)	Salinas	97.12	SAE with Spectral-Spatial patches
Ma et al. (2016)	Indian Pines	96.70	Sparse Autoencoder
Tao et al. (2016)	PaviaU	96.85	Denoising Autoencoder
Li et al. (2017)	Indian Pines	97.00	Variational Autoencoder
Zhong et al. (2017)	Salinas	97.35	Contractive Autoencoder
Xu et al. (2020)	PaviaU, Salinas	98.02	AE + CNN Hybrid
Gao et al. (2021)	Indian Pines	97.83	Self-Supervised Autoencoder
DiffusionAAE (2025)	Indian Pines	98.45	Diffusion-enhanced Adversarial Autoencoder
Hybrid AE+ GAN+ CNN (2025)	Salinas, Pavia Centre	98.60 (Salinas)	Autoencoder + GAN + CNN + Zero-shot Learning
SSDBN (2024)	Indian Pines, Salinas	98.25	Attention-based double-branch AE

Study	Dataset	Best Accuracy (%)	Remarks
3D-2D GAT (2024)	PaviaU	98.40	Hybrid AE with Graph Attention
Our Work	Pavia Centre (PaviaC)	**98.848**	Best performance using SGD optimizer
Our Work	Pavia University (PaviaU)	**96.012**	Stable performance with SGD optimizer
Our Work	Botswana	**99.815**	Highest accuracy with Adagrad optimizer
Our Work	KSC	**97.621**	Strong performance with SGD optimizer
Our Work	Indian Pines	**96.878**	Good result with Adagrad optimizer

6 Conclusion and Future Scope

In this work, a novel deep learning model combining an autoencoder with a classifier was proposed for hyperspectral image classification. The architecture leverages the strengths of both unsupervised feature learning through reconstruction and supervised classification through discriminative training. By extracting compressed spectral-spatial features via the encoder and simultaneously reconstructing the input and predicting class labels, the model achieves robust generalization and effective feature representation. Experimental results across multiple benchmark hyperspectral datasets demonstrate that the proposed HyperspectralAE model achieves competitive classification accuracy while maintaining reconstruction quality, thereby confirming the effectiveness of the joint learning approach. This architecture is particularly well-suited for scenarios with limited labeled data, a common challenge in hyperspectral remote sensing applications.

Although the proposed HyperspectralAE shows promising results, several potential directions remain open for future exploration. First, the incorporation of attention mechanisms could further enhance feature extraction by dynamically focusing on the most relevant spectral and spatial regions. Second, expanding the model towards semi-supervised or self-supervised frameworks could significantly reduce the dependency on labeled samples, which are expensive to obtain in hyperspectral imaging. Additionally, integrating graph convolutional networks (GCNs) to model long-range spatial relationships may improve the classification of spatially dispersed classes. Finally, deploying the model for real-time hyperspectral applications, such as precision agriculture or disaster monitoring, will require optimization strategies for reducing model complexity without compromising accuracy, paving the way for efficient and scalable deployment on resource-constrained platforms.

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